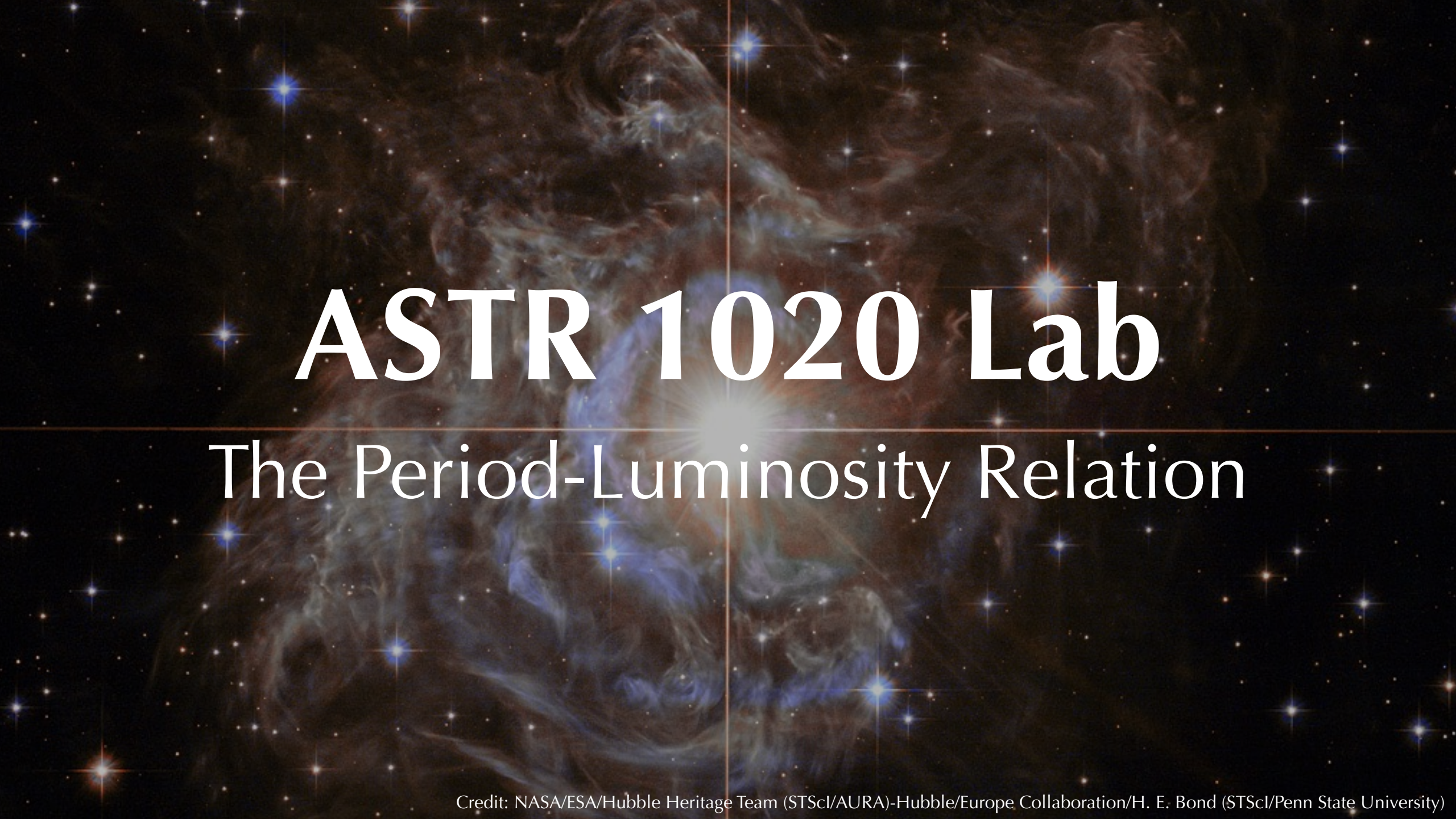


ASTR 1020 Lab

The Period-Luminosity Relation

Credit: NASA/ESA/Hubble Heritage Team (STScI/AURA)-Hubble/Europe Collaboration/H. E. Bond (STScI/Penn State University)

Grading

- All labs will be graded out of **10 points***

Points per question	Description
1.0	A correct answer **with** units **and** work shown. Answers that don't require work will be **graded on completion**
0.8	A correct answer **without** units or work shown
0.6	An incorrect answer **with** units **and** work shown
0.4	An incorrect answer **without** units or work shown
0.2	Some work shown **without** an answer
0.0	Not Attempted

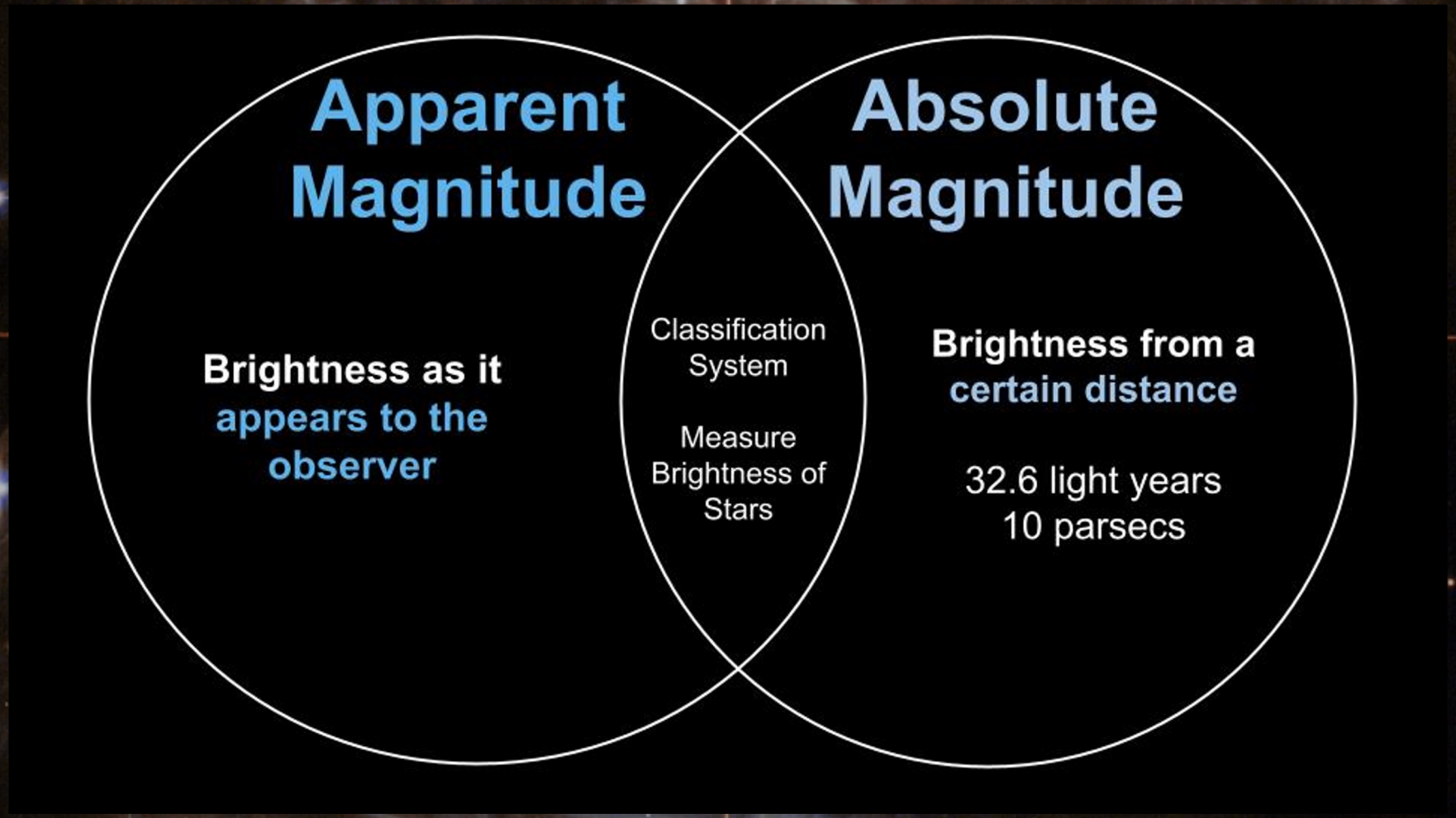
Lab Schedule

Lab	Dates	Topic
~~Lab 1~~	~~September 14~~	~~Excel & Data Analysis~~
~~LAB 2~~	~~September 21~~	~~Sun~~
~~LAB 3~~	~~September 28~~	~~Brightness & Luminosity~~
~~LAB 4~~	~~October 5~~	~~Spectra~~
LAB 5	October 12	Period–Luminosity
LAB 6	October 19	Hubble
LAB 7	October 26	Stellar Evolution
LAB 8	November 2	Galaxies
LAB 9	November 9	Star Clusters & Supernovae
LAB 10	November 16	Black Holes

Things You Need To Know For Lab 5

- Using excel
- Difference between apparent and absolute magnitude
- Distance modulus equation

Apparent vs Absolute Magnitude



Credit: Learn the Sky

Distance Modulus

- **Solve for distance** if absolute & apparent magnitudes are known

$$M = m + 5 - 5*\log(D)$$

Absolute Magnitude

Apparent Magnitude

Distance (in **parsecs**)